

Submission — Lab 5.2:

AstraNotes Requirements-to-UML Traceability Matrix

1. Traceability Matrix

Req ID	Requirement (short)	Class / Object Evidence	Use Case / Activity Evidence	Deployment Evidence	Status	Gap Note
FR-1	Create text note with title and markdown body.	Note class with <code>id</code> , <code>title</code> , <code>body</code> , <code>kind</code> , <code>created_at</code> , <code>updated_at</code> ; NoteService.create() .	Use case “Create note (FR-1)”; activity diagram (specialized for SecureNote) shows the same NoteService.create() entry path.	App → Services → LocalSQLiteRepository → SQLite store on User Device.	Fully Traced	—
FR-4	Mark note as private (SecureNote).	SecureNote (extends Note) with <code>password_hash</code> , <code>encrypted_body</code> ; EncryptionService .	Use case “Mark note as private (FR-4, SEC-1)”; activity diagram steps F–G perform key derivation and encryption.	Services block on User Device; ciphertext sits in the local SQLite store.	Fully Traced	—
FR-6	Maintain version history; allow restore.	VersionHistory class; VersionHistoryService with <code>snapshot</code> / <code>list</code> / <code>restore</code> .	Use case “View / restore version history (FR-6)”; activity diagram step I appends a snapshot on every save.	Stored alongside notes in <code>~/AstraNotes/store.db</code> .	Fully Traced	—
FR-7	Cloud Sync across devices.	SyncService , RemoteRepository (implements NoteRepository).	Use cases “Enable Cloud Sync (FR-7)” and “Resolve sync conflict (FR-7)”; no activity diagram for the sync workflow.	Optional Cloud zone; SyncService ↔ SyncServer over TLS ; server holds ciphertext only.	Partially Traced	No activity diagram exists for the sync + conflict-resolution workflow. Recommend adding one before implementation begins.
FR-8	Search notes by title/body, ranked by recency.	No dedicated SearchService class; search is implied as a method on NoteRepository .	Use case “Search notes (FR-8)” exists; no activity diagram for the search path.	Deployment shows an “FTS5 index for FR-8” annotation on the SQLite store.	Weakly Traced	Class diagram does not show how SecureNote bodies are excluded from search until unlocked. Need an explicit SearchService (or method on NoteService) and an activity diagram covering the locked-result branch.

Traceability matrix (continued)

Req ID	Requirement (short)	Class / Object Evidence	Use Case / Activity Evidence	Deployment Evidence	Status	Gap Note
NFR-1	Open / search 10k notes < 100 ms p95.	Implied by <code>LocalSQLiteRepository</code> + FTS5 annotation in deployment.	Not addressed by any use case or activity diagram.	Deployment annotates an FTS5 index on the SQLite store.	Weakly Traced	NFR-1 is a performance constraint that no UML view explicitly demonstrates. Sprint-Zero spike S0-11 covers measurement; matrix flags this as evidence-deferred-to-spike.
SEC-1	Encrypt SecureNote bodies at rest.	<code>EncryptionService</code> ; <code>SecureNote.encrypted_body</code> ; <code>password_hash</code> field.	Activity diagram step G (“Encrypt body → <code>encrypted_body</code> ”) and step H (“Repository writes ciphertext only — SEC-1”).	Deployment annotation: “Sync Server stores ciphertext only — SEC-1.”	Fully Traced	—
SEC-4	Append-only audit log.	<code>AuditLogEntry</code> class; <code>NoteService</code> → <code>AuditLogEntry</code> : writes association.	Activity diagram step J (“Append <code>AuditLogEntry</code> ‘note.created’ SEC-4”).	Deployment shows “Append-only audit log” as a separate file on the User Device.	Fully Traced	—

2. Traceability Metrics

- **Total requirements reviewed:** 8 (the most important ones across FR / NFR / SEC).
- **Fully Traced:** 5 (FR-1, FR-4, FR-6, SEC-1, SEC-4).
- **Partially Traced:** 1 (FR-7).
- **Weakly Traced:** 2 (FR-8, NFR-1).
- **Not Traced:** 0.
- **Major UML elements without a clear requirement reason to exist:** 1 candidate — the `ValidationService` class is currently invoked inside the activity diagram but is not directly referenced by any requirement ID; it supports SEC-2 (graceful failure) implicitly. Either link it explicitly to SEC-2 in the next revision, or replace it with inline checks inside `NoteService`.

3. Gap Analysis

The package is structurally strong on the security and version-history requirements — those are the elements where the Charter is most opinionated and where I spent the most refinement effort. The gaps cluster around three areas, in order of how much they should change before implementation begins:

1. **FR-7 (Cloud Sync)** has all the structural pieces (`SyncService`, `RemoteRepository`, optional Cloud zone) but no activity diagram for the conflict-resolution path that the requirement explicitly demands (“user is prompted rather than silent merge”). Without a visual workflow, the implementation will reinvent the conflict policy. **Action:** add a sync activity diagram before sprint planning for US-5.
2. **FR-8 (Search)** is the weakest link — the class diagram has no `SearchService` and the locked-SecureNote handling rule from the refined requirement is not visible anywhere in UML. **Action:** add a thin `SearchService` (or document that search lives on `NoteRepository`) and capture the locked-result branch in either the class diagram notes or a small activity diagram. This also tightens the NFR-1 story.
3. **NFR-1 (performance)** is, by nature, hard to express in UML, but the deployment-level FTS5 annotation is currently the only evidence. **Action:** rely on Sprint-Zero spike S0-11 for measured evidence and add a one-line note to the class diagram documenting that `NoteRepository.list()` and search use the FTS5 index.

The single suspect element (`ValidationService`) should be explicitly justified by SEC-2 in the next revision rather than left implicit; small accretions like this are how design packages drift from “every element has a requirement reason to exist.”

How AI helped

I used Copilot Chat to draft an initial matrix with all 14 requirement IDs and let it propose *Fully* / *Partially* / *Weakly* labels. I revised the matrix because Copilot was too strict on some and overly lenient on the others.